

Week 2 Report

HMM Architecture and Training Algorithms

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1. Markov chains
2. Hidden Markov Model architecture
3. HMM probability model and assumptions
4. Forward–Backward algorithm
5. Baum–Welch training

Markov Chain

A **Markov chain** $\{Z_t\}_{t \in [1, T]}$ is a stochastic process describing a sequence of states

$$Z_1, Z_2, \dots, Z_T$$

where the current state only depend on the previous state (first-order).

Markov Property

$$P(Z_t \mid Z_1, \dots, Z_{t-1}) = P(Z_t \mid Z_{t-1}).$$

Key Idea

Predictions can be made regarding future outcomes based solely on its present state and such predictions are just as good as the ones that could be made knowing the process's full history.

Transition Probabilities

The dynamics of a Markov chain are described by a **transition matrix**

$$A_{ij} = P(Z_t = j \mid Z_{t-1} = i).$$

For instance, for a three-state model,

$$A = \begin{bmatrix} 0.90 & 0.07 & 0.03 \\ 0.10 & 0.80 & 0.10 \\ 0.05 & 0.15 & 0.80 \end{bmatrix}.$$

Each row must satisfy

$$\sum_{j=1}^K A_{ij} = 1.$$

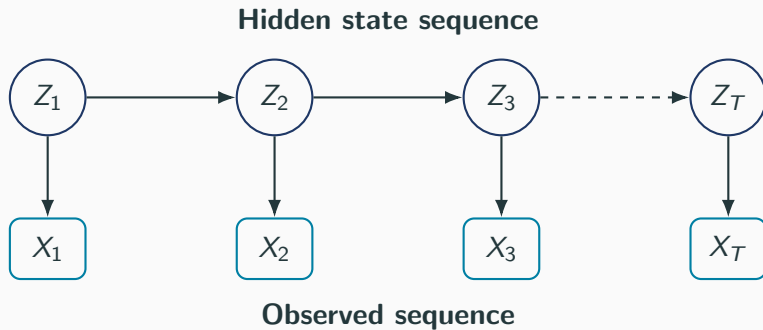
From Markov Chains to Hidden Markov Models

In typical Time Series analysis, we model each data point as a Random Variable that are directly time-dependent.

In a **Hidden Markov Model (HMM)**, the temporal dependence is hidden in unobserved states that form a Markov chain, and we assume the data points are directly generated by these hidden states.

In our application: assume all market features are directly specified by the regime they belong to, and only the current regime have influence on the probabilities of the immediate future regime.

Hidden Markov Model Architecture



$$Z_t \longrightarrow Z_{t+1} \quad \text{and} \quad Z_t \longrightarrow X_t.$$

Components of an HMM

A standard HMM can be represented as $\lambda = (\pi, A, \theta)$

1. Initial-State Distribution

$$\pi_i = P(Z_1 = i)$$

determines the probability of starting in state i .

2. Transition Matrix

$$A_{ij} = P(Z_t = j \mid Z_{t-1} = i)$$

determines how hidden states evolve over time.

3. Emission Distribution

$$p(X_t \mid Z_t = i; \theta_i)$$

describes how observations are generated from each hidden state.

Gaussian Emission Model

For continuous data, a classical HMM commonly assumes

$$X_t \mid Z_t = k \sim \mathcal{N}(\mu_k, \Sigma_k).$$

Therefore, each hidden state has its own

$$\mu_k \quad \text{and} \quad \Sigma_k.$$

Important for Our Project

The emission model $p(X_t \mid Z_t)$ is the component we eventually want to make more flexible.

Conditional Independence Assumptions

An HMM relies on two important assumptions.

1. First-Order Markov Assumption

$$P(Z_t \mid Z_1, \dots, Z_{t-1}) = P(Z_t \mid Z_{t-1}).$$

The current hidden state depends only on the previous hidden state.

2. Emission Independence Assumption

$$P(X_t \mid Z_{1:t}, X_{1:t-1}) = P(X_t \mid Z_t).$$

Once Z_t is known, the observation X_t depends only on the current state.

Joint Probability of an HMM

Using the HMM assumptions, the joint probability factorizes as

$$P(X_{1:T}, Z_{1:T}) = P(Z_1) \prod_{t=2}^T P(Z_t \mid Z_{t-1}) \prod_{t=1}^T P(X_t \mid Z_t).$$

Why This Matters

This factorization makes efficient dynamic-programming algorithms possible for HMM inference and training.

Three Classical HMM Problems

1. Likelihood

$$P(X_{1:T} \mid \lambda)$$

How likely is the observed sequence under the model?

2. Decoding

$$Z_{1:T}^* = \arg \max_{Z_{1:T}} P(Z_{1:T} \mid X_{1:T})$$

What's the most probable sequence given the model?

3. Learning

$$\pi, \quad A, \quad \theta$$

How do we estimate these quantities when the hidden states are not observed?

Algorithms for the Three Problems

Problem	Goal	Algorithm
Likelihood	Compute $P(X_{1:T})$	Forward algorithm
Decoding	Estimate most likely state sequence	Viterbi algorithm
Learning	Estimate HMM parameters	Baum–Welch / EM

Baum–Welch is the **Expectation-Maximization** Algorithm specialized for HMMs.

Forward Algorithm

Define the forward probability

$$\alpha_t(i) = p(Z_t = i | X_{1:t}).$$

Initialization

$$\alpha_1(i) = \frac{p(X_1 | Z_1 = i)p(Z_1 = i)}{\sum_j p(X_1 | Z_1 = j)p(Z_1 = j)} = \frac{p(X_1 | Z_1 = i)\pi_i}{\sum_j p(X_1 | Z_1 = j)p(Z_1 = j)}.$$

Recursion

$$\alpha_t(j) = p(X_t | Z_t = j) \sum_{i=1}^K \alpha_{t-1}(i) A_{ij}.$$

Backward Algorithm

Define the backward probability

$$\beta_t(i) = P(X_{t+1:T} \mid Z_t = i).$$

Initialization

$$\beta_T(i) = 1.$$

Recursion

$$\beta_t(i) = \sum_{j=1}^K A_{ij} p(X_{t+1} \mid Z_{t+1} = j) \beta_{t+1}(j).$$

Baum–Welch: E-Step

The E-step computes posterior probabilities using the current model parameters.

The probability of being in state i at time t is

$$\gamma_t(i) = P(Z_t = i \mid X_{1:T}).$$

Using forward and backward probabilities,

$$\gamma_t(i) = \frac{\alpha_t(i)\beta_t(i)}{\sum_{j=1}^K \alpha_t(j)\beta_t(j)}.$$

Interpretation

$\gamma_t(i)$ is the posterior probability that the hidden state at time t is i , given the entire observed training sequence.

Baum–Welch: Transition Probabilities

We also calculate

$$\xi_t(i, j) = P(Z_t = i, Z_{t+1} = j \mid X_{1:T}).$$

Using forward and backward probabilities,

$$\xi_t(i, j) = \frac{\alpha_t(i) A_{ij} p(X_{t+1} \mid Z_{t+1} = j) \beta_{t+1}(j)}{P(X_{1:T})}.$$

Interpretation

$\xi_t(i, j)$ measures the posterior evidence that the process transitioned from state i to state j between times t and $t + 1$.

After estimating the hidden states, we update the model parameters.

Initial-state probabilities

$$\pi_i^{\text{new}} = \gamma_1(i).$$

Transition probabilities

$$A_{ij}^{\text{new}} = \frac{\sum_{t=1}^{T-1} \xi_t(i, j)}{\sum_{t=1}^{T-1} \gamma_t(i)}.$$

For a Gaussian-emission HMM,

$$X_t \mid Z_t = i \sim \mathcal{N}(\mu_i, \Sigma_i).$$

The updated mean is

$$\mu_i^{\text{new}} = \frac{\sum_{t=1}^T \gamma_t(i) X_t}{\sum_{t=1}^T \gamma_t(i)}.$$

The covariance is

$$\Sigma_i^{\text{new}} = \frac{\sum_{t=1}^T \gamma_t(i) (X_t - \mu_i^{\text{new}})(X_t - \mu_i^{\text{new}})^T}{\sum_{t=1}^T \gamma_t(i)}.$$

We cannot directly maximize the joint likelihood $p(x_{1:T}, z_{1:T} | \theta)$ since we don't actually know $z_{1:T}$. The parameters are updated by using MLEs of the **expected log joint likelihood**:

$$Q(\theta | \theta^{\text{old}}) = \mathbb{E}_{z|x, \theta^{\text{old}}} [\log p(x_{1:T}, z_{1:T} | \theta)]$$

$$\theta^{\text{new}} = \operatorname{argmax}_{\theta} Q(\theta | \theta^{\text{old}})$$

Complete Baum–Welch Algorithm

Algorithm 1: Baum–Welch Training for a Gaussian HMM

Input: Observation sequence $x_{1:T}$, number of hidden states K

Output: Learned HMM parameters θ

```
1 Initialize  $\theta^{(0)} = (\pi, A, \mu, \sigma^2)$ ;  
2 repeat  
    // E-Step  
3     Compute  $\alpha_t(i), \beta_t(i), \gamma_t(i), \xi_t(i, j) \forall i, j$  in  $[1, 2, \dots, K]$ ;  
    // M-Step  
4     Update initial-state probabilities  $\pi$ ;  
5     Update transition matrix  $A$  using  $\gamma$  and  $\xi$ ;  
6     Update Gaussian means  $\mu_i$  using  $\gamma$ ;  
7     Update Gaussian variances  $\sigma_i^2$  using  $\gamma$ ;  
8 until observed data log-likelihood  $\log p(x_{1:T}|\theta)$  converges;  
9 return trained parameters  $\theta$ ;
```

Limitations of Gaussian Emissions

A standard Gaussian HMM assumes

$$X_t \mid Z_t = k \sim \mathcal{N}(\mu_k, \Sigma_k).$$

However, financial return distributions may exhibit

- heavy tails,
- skewness,
- multimodality,
- nonlinear dependence between variables.

A single Gaussian distribution may therefore be too restrictive to accurately model the observations generated by each market regime.

Limitation of Gaussian Emissions

Notice that

$$\alpha_t(i) = p(x_t | z_t = i) \sum_j A_{ij} \alpha_{t-1}(j)$$

$$\beta_t(i) = \sum_j A_{ij} p(x_{t+1} | z_{t+1} = j) \beta_{t+1}(j)$$

And these quantities are the building blocks of **every** single parameter estimation in the algorithm.

Implication

The emission probability has large influence to the model learning process.

- Preprocess data from CRSP
- Implement a classical Gaussian HMM for benchmark.